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Smart Municipal Reporting Platform

FixCity – Mobile Application

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Abstract

In the context of rapid urbanization and the national push for digital transformation in Egypt, the disconnect between citizens and municipal authorities remains a critical barrier to effective public service delivery. Traditional methods of reporting urban infrastructure failures—such as potholes, broken lighting, and sanitation issues—are characterized by inefficiency, lack of transparency, and low public engagement.

This project presents FixCity, a native mobile application built upon Flutter and Supabase, and enhanced with Google Gemini AI for intelligent category classification. The system democratizes the reporting process, allowing citizens to submit geo-tagged, visually verified problem reports. The project was accepted into the Gemini for Graduation Projects program (Cohort III) by Gemini Africa, in partnership with the Academy of Scientific Research and Technology.

By applying Business Process Digitalization principles, FixCity transforms unstructured citizen complaints into structured, actionable data for municipal decision-makers. The research demonstrates that a Minimum Viable Product (MVP) approach can deliver a cost-effective, scalable solution that enhances operational efficiency and fosters civic trust. This report details the comprehensive system analysis, business modeling, and technical implementation required to deploy such a solution within the resource constraints of small-to-medium Egyptian municipalities.

Keywords: *Civic Technology, Digital Transformation, Smart Cities, Urban Reporting, E-Government, Business Process Improvement, Minimum Viable Product (MVP), Flutter, Supabase, Google Gemini AI.*

1. Introduction

1.1 General Background

The concept of the "Smart City" has transitioned from a futuristic ideal to a necessary framework for managing modern urbanization. As population densities in metropolitan areas increase, the strain on traditional public infrastructure—roads, lighting, sanitation, and utilities—grows exponentially. In the context of the Arab Republic of Egypt, this challenge is compounded by rapid urban expansion and the legacy of manual administrative systems.

Digital Transformation (DX) has emerged as a primary national objective, aiming to modernize government services and enhance the quality of life for citizens. However, while high-level government services (such as tax filing or vehicle licensing) have seen significant digitization, the "last mile" of municipal management—specifically the maintenance of local urban infrastructure—often remains reliant on outdated, paper-based workflows.

This disconnect creates a "Digital Divide" between the government's strategic goals and the citizen's daily reality. Civic Technology, defined as digital tools designed to enhance citizen engagement, offers a bridge. By leveraging accessible mobile technologies, municipalities can decentralize the monitoring of infrastructure, effectively turning every citizen with a smartphone into a sensor for the city.

This project, "FixCity," proposes a mobile-based solution to this specific problem. It is not merely a software application but a Business Process Improvement (BPI) initiative designed to streamline the workflow of problem reporting, tracking, and resolution.

1.2 Problem Statement

The core problem addressed by this research is the inefficiency of current communication channels between urban residents and municipal authorities in Egyptian cities.

1.2.1 The Inaccessibility of Reporting

Currently, residents facing common infrastructure failures—such as dangerous potholes, uncollected waste, or broken streetlights—have limited options for reporting these issues. The primary mechanisms are:

- Physical Visits: Requiring citizens to travel to municipal offices during working hours.
- Phone Calls: Often routed to general switchboards without dedicated tracking systems.
- Informal Complaints: Lodged through personal connections or local representatives.

These methods present a high barrier to entry for the average citizen. As a result, data collected during the preliminary research for this project indicates that approximately 70% of urban problems go unreported simply due to the difficulty of the reporting process. This "silent majority" of unreported issues leads to the gradual degradation of urban environments.

1.2.2 Operational Inefficiency

From the municipal perspective, the reliance on manual inputs results in severe operational bottlenecks. When complaints are received via phone or paper, they lack standardization. A

citizen might report a "hole in the road" without specifying the exact location or severity. Municipal workers are then forced to spend valuable time on administrative triage—clarifying details, transcribing forms, and manually routing tasks—rather than focusing on problem resolution. This manual handling frequently leads to lost information and delays.

1.2.3 The Transparency Gap

Perhaps the most damaging aspect of the current system is the lack of feedback loops. Citizens who do take the time to report an issue rarely receive confirmation that their report was received, let alone updates on its resolution status. This lack of transparency fosters civic apathy and erodes trust between the public and service providers.

1.3 Project Motivation

The motivation for developing FixCity stems from the urgent need to apply Business Process Digitalization to the municipal sector.

- **Improving Public Safety & Quality of Life:** Unresolved infrastructure issues are not mere inconveniences; they pose genuine safety risks. A deep pothole can damage vehicles or cause accidents; unlit streets can encourage crime; accumulated trash poses health hazards. Accelerating the reporting loop directly contributes to a safer urban environment.
- **Resource Optimization:** By shifting from unstructured manual complaints to structured digital data (e.g., GPS coordinates, categorized drop-downs), municipalities can optimize their resource allocation. Workers can group repairs by location rather than responding to ad-hoc phone calls.
- **Democratizing Access:** Smartphone penetration in Egypt is high across various socioeconomic demographics. A mobile-first platform removes the socio-economic barriers of "knowing the right person" to get a pothole fixed, allowing any citizen with internet access to participate in urban maintenance.

1.4 Research Objectives

This project aims to design, develop, and validate a Minimum Viable Product (MVP) that demonstrates the viability of digital urban reporting in the Egyptian context.

1.4.1 Primary Objectives

- **Develop a Functional Mobile Application:** To engineer a native mobile application using Flutter that allows users to submit standardized reports including location data, problem categories, and photographic evidence.
- **Implement a Cloud Backend:** To utilize Supabase as a scalable backend infrastructure to manage user authentication, data storage, and real-time status updates without complex server maintenance.
- **Establish a Feedback Loop:** To create an automated notification system that informs citizens of the status of their reports (New → In Progress → Resolved), thereby ensuring transparency.

1.4.2 Secondary Objectives

- **Validate User Acceptance:** To conduct user surveys (sample size n=150) confirming that the proposed digital solution aligns with citizen preferences for mobile engagement.
- **Analyze Business Viability:** To provide a cost structure analysis proving that such a system can be implemented with a startup capital as low as \$350, making it accessible for small municipalities.

1.5 Alignment with UN Sustainable Development Goals (SDGs)

Figure 1: UN Sustainable Development Goals (SDGs)



This project extends beyond technical implementation; it is designed to align with and contribute to the United Nations Sustainable Development Goals (SDGs). By modernizing urban management in Egypt, FixCity directly addresses critical challenges related to health, infrastructure, and community resilience. The platform actively contributes to the following four goals:

SDG 3: Good Health and Well-being

Poorly managed urban environments act as catalysts for public health crises. Accumulated garbage, stagnant water due to broken drainage, and open sewage systems are primary breeding grounds for disease vectors. By significantly accelerating the reporting and resolution cycle of

these environmental hazards, FixCity mitigates public health risks and contributes to cleaner, healthier living conditions for urban residents.

SDG 6: Clean Water and Sanitation

Water scarcity is a pressing national issue. Traditional, slow reporting mechanisms for water main bursts or sewage overflows result in significant resource wastage and sanitation hazards. FixCity provides an immediate channel for reporting water infrastructure failures, enabling faster municipal response times to conserve water resources and ensure proper sanitation management.

SDG 9: Industry, Innovation, and Infrastructure

This project represents a direct application of digital innovation to modernize legacy infrastructure management. By replacing outdated, reactive paper-based workflows with a proactive, data-driven digital platform, FixCity enhances the efficiency, resilience, and sustainability of critical urban infrastructure.

SDG 11: Sustainable Cities and Communities

This is the primary overarching goal of the FixCity initiative. The platform democratizes urban governance by empowering citizens to actively participate in maintaining their environment. Furthermore, by streamlining waste and infrastructure management, it directly contributes to reducing the adverse per capita environmental impact of cities, fostering safer, more inclusive, and resilient urban communities.

1.6 Project Scope

Defining the boundaries of the project is essential to manage expectations and ensure the successful delivery of the Minimum Viable Product (MVP). The scope is delineated across three dimensions: Domain, Geography, and Functionality.

1.6.1 Domain Scope

The FixCity project is situated at the intersection of Civic Technology and Business Information Systems. Specifically, it addresses the domain of Digital Government Services, focusing on "Citizen Engagement Platforms." It utilizes principles of Business Process Digitalization to transform manual municipal workflows into automated digital systems. The project does not aim to reinvent municipal policy but rather to provide the technological infrastructure to execute existing policies more efficiently.

1.6.2 Geographic Scope

While the underlying technology is location-agnostic, the system design is tailored specifically for the Egyptian urban context.

- **Interface Localization:** The user interface supports Arabic as the primary language to ensure accessibility for local residents, with English available for administrative components.
- **Infrastructure Typology:** The reporting categories (e.g., "Potholes," "Trash Accumulation") are selected based on the most frequent complaints cited in Egyptian municipal studies.

- **Implementation Level:** The current scope is limited to a conceptual implementation for a single district or small municipality rather than a nationwide deployment.

1.6.3 Functional Scope

The system is designed to support the complete lifecycle of a non-emergency infrastructure report.

- **In-Scope:** Mobile reporting interface (Android), photo uploads via camera/gallery, geolocation tagging via device sensors, administrative dashboard, and real-time status tracking.
- **Out-of-Scope:** Emergency services dispatch (police/fire/ambulance), integration with payment gateways (for fines or fees), or complex GIS mapping for urban planning.

1.7 Project Limitations

It is critical to acknowledge the constraints under which this research was conducted. These limitations frame the interpretation of the results and the functionality of the final artifact.

- **Academic Simulation:** The integration with municipal workflows is simulated for academic purposes. A real-world deployment would require complex API integrations with legacy government systems and bureaucratic approval processes that are beyond the scope of a graduation project.
- **MVP Functionality:** The system represents a Minimum Viable Product (MVP). Advanced features such as AI-driven image recognition for automated severity assessment or blockchain for immutable audit logs were excluded to ensure the project remained feasible within the development timeline.
- **Digital Literacy Assumption:** The solution presupposes a baseline level of digital literacy among users (specifically, the ability to operate a smartphone). While mobile penetration is high, this reliance may inadvertently exclude older or economically disadvantaged demographics.

1.8 Definitions of Key Terms

- **Civic Technology (Civic Tech):** Digital tools and platforms specifically designed to enhance citizen engagement with government services and improve the delivery of public goods. FixCity is a "Citizen-to-Government" (C2G) Civic Tech application.
- **Business Process Digitalization:** The transformation of manual, paper-based workflows into digital systems to improve efficiency, accuracy, and accessibility. This differs from simple "digitization" (scanning papers) as it fundamentally reorganizes the process flow.
- **Minimum Viable Product (MVP):** A development technique where a new product is built with sufficient features to satisfy early adopters while collecting feedback for future development. This iterative approach reduces the risk of building features that users do not need.

1.9 Report Organization

The remainder of this report is organized as follows:

- Chapter 2: Background & Literature Review establishes the theoretical framework, reviewing the history of digital government in Egypt and analyzing existing solutions like FixMyStreet.
- Chapter 3: Business Model & Marketing Plan details the operational strategy, including the Business Model Canvas, cost structure analysis, and a phased implementation roadmap.
- Chapter 4: System Analysis documents the requirements engineering process, including stakeholder interviews, functional specifications, and data modeling.
- Chapter 5: System Design & Implementation details the technical architecture, user interface design, and AI integration layer of the FixCity platform.

2. Background & Literature Review

2.1 Introduction

The development of FixCity relies on established principles of mobile application development and public administration. This chapter provides the theoretical underpinning for the project. It begins by exploring the technological foundations—specifically the selection of the Flutter/Supabase framework and database strategies. It then moves to a critical literature review, surveying existing solutions such as international platforms (FixMyStreet) and local informal channels (social media groups) to identify the specific market gaps this project aims to address.

2.2 Theoretical Foundation

2.2.1 Mobile Application Development Framework

The technical architecture of FixCity is built upon the Flutter/Supabase framework.

- **Selection Justification:** Flutter was selected for its cross-platform capability, compiling to native Android code from a single Dart codebase. Supabase was selected as the backend for its production-ready PostgreSQL database, built-in authentication, real-time subscriptions, and file storage buckets — eliminating the need for dedicated server management.
- **Scalability:** While used here for an MVP, Flutter/Supabase is industry-proven and capable of scaling to handle high traffic loads, ensuring the project's logic is future-proof.

2.2.2 Database Management Systems

Effective data management is the backbone of any reporting system. FixCity employs PostgreSQL as its primary database engine, hosted on the Supabase cloud platform.

PostgreSQL is an object-relational database management system (ORDBMS) renowned for its reliability, data integrity, and support for complex queries. It is particularly well-suited for a live municipal environment handling concurrent user requests and geospatial data. Its use in both development and production environments eliminates compatibility issues, ensuring a seamless deployment pipeline.

2.2.3 User-Centered Design (UCD)

The project adheres to User-Centered Design principles, prioritizing the needs and constraints of the end-user.

- **Responsive Design:** Given that the majority of potential reports are expected to originate from mobile devices, the frontend utilizes Google's Material Design principles. This ensures the application is responsive, automatically adjusting layout and navigation elements to fit screen sizes ranging from small smartphones to large tablet screens, and performs smoothly on a wide range of devices utilizing native hardware features like GPS and camera.

2.3 Related Work and Competitive Analysis

To position FixCity effectively, it is necessary to analyze the current landscape of urban reporting tools.

2.3.1 International Benchmark: FixMyStreet (UK)

FixMyStreet is a pioneering platform launched in the United Kingdom that allows citizens to report street-level infrastructure problems on a map.

- **Strengths:** It has achieved widespread public adoption and offers seamless integration with local council back-office systems. It sets the standard for transparency and usability in the Civic Tech space.
- **Weaknesses:** The platform is technically complex and resource-intensive. Implementing a similar system in Egypt would require significant IT infrastructure upgrades at the municipal level, which is often financially unfeasible for smaller districts.

2.3.2 Local Alternative: Municipal Contact Forms

Various Egyptian governorates have digitized their complaints process by adding web forms to their official portals.

- **Strengths:** These channels carry the authority of the official government.
- **Weaknesses:** They typically function as simple email relays. They lack tracking capabilities, meaning citizens receive no feedback. Furthermore, the user experience is often poor, with non-responsive designs that are difficult to use on mobile devices.

2.3.3 Informal Alternative: Social Media Groups

In the absence of effective official tools, citizens have turned to Facebook groups and WhatsApp communities to report issues.

- **Strengths:** High accessibility and immediate community validation.
- **Weaknesses:** The data is unstructured and informal. Reports are often buried under other content, and there is no official mechanism for the municipality to acknowledge or resolve these "posts".

2.3.4 Comparative Analysis Matrix

Table 1: System Comparison

Aspect	FixCity	Municipal Contact Forms	Social Media Groups	FixMyStreet (UK)
Problem Focus	Urban infrastructure	General inquiries	Community discussions	Street problems
Workflow	Basic tracking system	Email forwarding	No formal workflow	Advanced workflow
Technical Level	Native mobile application	Basic HTML forms	Platform-dependent	Complex system
Accessibility	Mobile-responsive	Variable	High	High
Cost	Low implementation	Low	Free	High
Customization	Flexible	Limited	None	Extensive
Official Response	Simulated integration	Direct email	Informal	Formal integration
Transparency	Status tracking	Limited	High	Comprehensive

This analysis confirms that FixCity fills a critical gap: it offers the structured data and tracking of an advanced system like FixMyStreet but retains the low cost and simplicity required for the Egyptian context.

2.4 Technological Foundation

The selection of the technology stack for FixCity was driven by the need for high performance, cross-platform compatibility, and rapid development cycles.

2.4.1 Frontend Framework: Flutter

The mobile application is developed using Flutter, an open-source UI software development kit created by Google.

- **Justification:** Unlike web-based solutions that rely on browser rendering, Flutter compiles to native machine code for both Android and iOS from a single codebase (Dart). This ensures high performance (60 FPS rendering) and provides a consistent, native user experience across different devices without the cost of maintaining two separate development teams.

2.4.2 Backend Infrastructure: Supabase

For the backend, the project utilizes Supabase, an open-source Backend-as-a-Service (BaaS) platform.

- **Justification:** Supabase provides a production-ready PostgreSQL database, instant APIs, and robust authentication out of the box. This eliminates the need to manually configure and maintain a dedicated server, allowing the development team to focus on application

logic rather than infrastructure management. It also supports real-time subscriptions, enabling instant status updates for users.

2.4.3 Map Services: CartoDB / OpenStreetMap

To handle geolocation and mapping without incurring the high costs of Google Maps API, the system integrates CartoDB tile layers via the `flutter_map` package. CartoDB was specifically chosen after OpenStreetMap's tile servers blocked the application due to usage policy restrictions, making CartoDB the production-ready, unrestricted alternative.

- Justification: This open-source alternative provides sufficient accuracy for urban reporting while keeping the project's operational costs near zero, aligning with the budget constraints of small municipalities.

2.5 Summary

The literature review confirms a distinct gap in the current market: a lack of affordable, structured, and transparent reporting tools for small-to-medium municipalities. While international solutions are too complex and local social media is too chaotic, FixCity leverages modern, lightweight technologies to offer a balanced, efficient solution. This theoretical and technical foundation paves the way for the specific Business Model discussions in the following chapter.

3. Business Model and Marketing Plan

3.1 Introduction

While FixCity is a technical solution, its long-term success depends on a sustainable business model that addresses the economic realities of Egyptian municipalities. This chapter transitions from the technical architecture to the operational strategy. It outlines how the platform creates, delivers, and captures value. Using the Business Model Canvas framework, we analyze the customer segments, key resources, and cost structures required to launch and maintain the system. Furthermore, this chapter presents a phased marketing plan designed to overcome the inertia often associated with adopting new technologies in the public sector.

3.2 Value Proposition

The core value of FixCity lies in its ability to solve the "last mile" problem of urban communication. The platform offers distinct value propositions for its two primary stakeholders: the citizens who report problems and the municipalities that resolve them.

3.2.1 For Citizens

For the urban resident, FixCity transforms the abstract frustration of living with poor infrastructure into actionable, trackable solutions.

- **Empowerment:** It provides a direct channel for residents who previously felt ignored by bureaucracy, giving them a voice in their local environment.
- **Convenience:** It eliminates the need for physical visits to government offices or navigating complex phone trees, allowing reports to be filed in under two minutes via smartphone.
- **Transparency:** Unlike traditional "black box" complaint methods, FixCity provides real-time status updates (e.g., "Received," "In Progress," "Resolved"), building trust that the system is working.

3.2.2 Channels

FixCity employs a multi-channel approach to reach these segments.

- **Direct Municipal Outreach:** The primary channel involves direct presentations to city council members.
- **App Store Distribution:** The platform is distributed via APK files or the Google Play Store, ensuring easy access for any Android user.
- **Community Partnerships:** Collaborating with local NGOs to facilitate grassroots adoption.

3.3 Business Model Canvas

Figure 2: FixCity Business Model

Key Partners	Key Activities	Value Proposition	Customer Relationships	Customer Segments
Technology Providers: - Supabase (Auth, PostgreSQL, Storage, Realtime) - Google Gemini AI (gemini-2.0-flash REST) - CartoDB / OpenStreetMap (flutter_map tiles) - Google Play Store Government Entities: - District Councils (pilot & domain authority) - Governors / District Heads (dedicated portal in app) NGOs & Community Groups: - Local orgs for citizen adoption & training Academic Institutions: - Universities for research & student testing	Platform Development: - Flutter multi-platform app (Android, iOS, Web) - Supabase backend (RLS, Realtime, RPC) - Gemini AI classification - Bilingual UI (EN + AR) Operations: - Admin & Governor portals - Municipal staff training - Marketing & outreach - Business dev w/ councils	For Citizens: - Report any infrastructure issue in under 2 minutes - Gemini AI auto-suggests the report category - Real-time status tracking: Pending → In Progress → Resolved - Unique 10-char code — no login required - Points & badges for every participation - Share reports via WhatsApp - Bilingual: English + Arabic For Municipalities: - Replaces manual phone calls and paperwork - Geo-tagged hotspot maps per district - Role-based access: Admin dashboard + Governor portal - Full audit trail per report - Before / After photo evidence for resolved reports - Lower cost vs. enterprise software	Automated Feedback: - In-app real-time notifications on every status change - Points awarded on submit and on admin resolution Technical Support: - Municipal staff training - Portal onboarding sessions Community Engagement: - WhatsApp deep-link sharing - Ping system for unresolved issues - Achievements & badges	Primary: - Small-to-Medium District Councils with limited IT budgets - Egyptian municipalities seeking digital tools Secondary: - Gated communities & compounds - Real-estate management companies End Users: - Urban residents aged 18-65 with smartphones - Bilingual users (EN + AR) - Citizens unable to visit government offices Built-in App Roles: - Citizen (public) - Municipal Admin - Governor / District Head
Cost Structure		Revenue Streams		
- Development: Flutter/Dart, VS Code, GitHub (~\$0 OSS tools; dev time is primary cost) - Infrastructure: Supabase free tier → Pro (~\$25/mo); CartoDB tiles (free for low traffic) - AI: Gemini API (free quota; pay-per-token beyond threshold) - Distribution: Google Play (25one – time); Apple(99/yr) - Marketing: Demo materials, campaigns, NGO outreach - Administrative: Legal, documentation, contracts		- Subscription Fees: Monthly SaaS to municipalities (100-500 / month per district) - Implementation Fees: One-time setup & customisation (1,000-3,000) - Training: Workshops for municipal staff (50-200 / person) - White-label Licensing: Custom version for gated communities or real-estate firms - Grants: Civic Tech funding (GIZ, USAID, Google.org)		

3.3.1 Customer Segments

The project targets a specific niche within the broader public sector market.

- **Primary Customers:** Small to medium-sized municipal governments (or district councils) in Egypt. These entities face high pressure to improve services but often possess the most limited IT budgets and technical resources.
- **Secondary Customers:** Community organizations, neighborhood associations (NGOs), and residential compounds looking to formalize problem reporting within their specific jurisdictions.
- **End Users:** The actual users of the software are urban residents (ages 18–65) experiencing infrastructure problems and seeking efficient reporting channels.

3.3.2 Customer Relationships

Building long-term relationships is critical for retention in the government sector.

- **Municipal Support:** The business model includes provisions for basic training, responsive technical assistance, and regular software updates to ensure security and stability.
- **Citizen Engagement:** For end-users, the relationship is maintained through transparent status notifications and continuous UX improvements based on feedback surveys.
- **Stakeholder Feedback:** Regular progress reporting to community organizations and periodic satisfaction surveys guide platform enhancements.

3.3.3 Key Partners

To minimize operational overhead, FixCity relies on strategic partnerships.

- **Technology Providers:** Supabase (for backend infrastructure), Google Gemini AI (for category classification), and CartoDB (for mapping services) are critical technology partners that lower operational costs.
- **Academic Institutions:** Universities play a dual role as research partners and potential pilot testing grounds.
- **NGOs & Community Groups:** Local organizations act as intermediaries to facilitate user testing and drive grassroots adoption among less digitally literate populations.
- **Government Entities:** Municipal departments offer domain expertise and opportunities for pilot implementation.

3.3.4 Key Activities and Resources

The core activities required to sustain the business include Platform Development (coding, feature enhancement), User Support (resolving technical issues), and Business Development (securing municipal contracts). Executing these activities requires specific resources:

- **Technical Infrastructure:** Reliable cloud hosting environments, development tools, and testing frameworks.
- **Human Capital:** A team combining software development expertise with project management skills and public administration domain knowledge.
- **Intellectual Property:** The proprietary source code, deployment methodologies, and best-practice documentation developed during the project.
- **Partnership Networks:** Established relationships with municipal contacts, academic advisors, and community organizations.

3.4 Financial Analysis

3.4.1 Cost Structure

The financial model for FixCity is designed to be "lean," minimizing upfront capital expenditure to make the solution affordable for smaller municipalities.

Table 3.1: Startup vs. Monthly Operational Costs

Table 2: Startup vs. Operational Costs

Cost Category	Startup Costs	Monthly Operational Costs	Description
Technology	\$25	\$5	App Store Developer Fees & Cloud Database Tiers.
Development	\$100	\$30	Development tools, software licenses, and ongoing code maintenance.
Administrative	\$50	\$20	Legal registration, basic communications, and documentation.
Marketing	\$50	\$25	Creation of demo materials and targeted digital outreach.
Total	\$350	\$120	Low entry barrier.

3.4.2 Revenue Streams

To ensure sustainability, the project proposes a diversified revenue model.

- **Municipal Licensing:** A subscription-based model charging \$100–\$500 monthly, scaled according to the size of the municipality and the feature set required.
- **Implementation Services:** One-time fees (\$1,000–\$3,000) for setting up the system, migrating existing data, and customizing the dashboard for specific local needs.
- **Training Programs:** Revenue generated by conducting training workshops for municipal staff to ensure digital literacy and system competence (\$50–\$200 per participant).
- **Grant Funding:** Pursuing non-dilutive funding from international organizations supporting civic technology and digital governance initiatives.

3.5 Marketing and Implementation Plan

3.5.1 Phase 1: Awareness Building (Months 1–3)

The initial phase focuses on establishing credibility. Activities include developing professional project documentation, presenting at local government technology forums, and establishing an online presence to showcase the prototype's capabilities.

3.5.2 Phase 2: Pilot Implementation (Months 4–6)

The goal of this phase is to secure a single municipal partner for a "Beta" deployment. This controlled environment will allow the team to document the implementation process, gather real-world performance metrics, and generate case studies that prove the system's value.

3.5.3 Phase 3: Regional Expansion (Months 7–12)

Armed with success stories from the pilot, the marketing focus shifts to regional expansion. This involves targeting adjacent municipalities (who often mimic successful neighbors) and developing a formal partnership program with community organizations to scale support.

3.5.4 Key Performance Indicators (KPIs)

Success will be measured against specific metrics within the first year:

- Adoption: Securing 2 municipal partnerships and achieving 5,000 active citizen users.
- Impact: Achieving a 30% reduction in problem resolution time compared to the pre-digital baseline.
- Satisfaction: Maintaining a user satisfaction score of 80% or higher to ensure long-term retention.

3.6 SWOT Analysis

To further validate the business model, a SWOT analysis was conducted to identify internal strengths and weaknesses, as well as external opportunities and threats.

- Strengths: Low implementation cost, agile MVP development, and a user-friendly interface tailored to the local language (Arabic).
- Weaknesses: As a new entrant, the project lacks the brand recognition of established competitors. The team also has limited initial resources for large-scale marketing.
- Opportunities: The Egyptian government's current push for "Digital Transformation" provides a favorable political climate. There is also a growing market for "Smart City" solutions in developing nations.
- Threats: Bureaucratic resistance to change is a significant barrier. Additionally, long procurement cycles in the public sector can strain cash flow for startup ventures.

4. System Analysis

4.1 Introduction

System analysis serves as the bridge between business needs and technical implementation. This chapter details the rigorous requirements engineering process undertaken for FixCity. It begins by outlining the data collection methodologies used to understand user needs, including quantitative surveys and qualitative stakeholder interviews. Following this, it presents a detailed breakdown of both Functional and Non-Functional Requirements, ensuring the system meets necessary performance and usability standards. Finally, the chapter defines the system's logical structure through Use Case analysis and Data Modeling, providing the blueprint for the design phase.

4.2 Requirements Elicitation

To ensure FixCity addresses real-world problems rather than theoretical ones, a user-centered requirements elicitation process was conducted.

4.2.1 Data Collection Methodology

- **Online Surveys:** A digital questionnaire was distributed to a diverse group of urban residents, garnering 167 responses. The survey assessed current reporting behaviors, identifying that the majority of users prefer mobile-based reporting over traditional phone or in-person methods.
- **Stakeholder Interviews:** Structured interviews were conducted with 8 municipal workers to understand the friction points in their daily workflows. These sessions revealed that "incomplete information" (e.g., reports without exact locations) was the primary cause of delay in problem resolution.
- **Process Mapping:** The team documented the existing manual workflows ("As-Is" process) to identify bottlenecks, such as the lack of feedback loops, which informed the "To-Be" digital process.

4.3 Functional Requirements

Functional requirements define the specific behaviors and functions the system must support. Based on the analysis, the following core functions were identified.

4.3.1 Citizen Module (Frontend)

- **FR1 – Problem Submission:** The system shall allow users to submit a new report containing mandatory fields: category (e.g., Pothole, Trash), precise location coordinates, and a text description of the issue.
- **FR2 – Media Evidence:** To validate reports, the system shall support optional photo uploads. The system must enforce validation rules, restricting file sizes to a maximum of 5MB to optimize storage.

- FR3 – Tracking Mechanism: Upon successful submission, the system shall automatically generate a unique, alphanumeric tracking ID. This ID allows users to query the status of their report later without needing to create a full account.

4.3.2 Administration Module (Backend Dashboard)

- FR4 – Dashboard Visualization: The system shall provide a secure dashboard for municipal workers to view a paginated list of all submitted problems.
- FR5 – Advanced Filtering: Admins must be able to filter reports based on specific criteria, such as Category (to route issues to the correct department), Status (New vs. Resolved), and Submission Date.
- FR6 – Workflow Management: The system shall enable authorized staff to update the status of a report. The defined status lifecycle is: New → In Progress → Resolved.

4.3.3 System Automation

- FR7 – Automated Notifications: To ensure transparency, the system shall trigger automated in-app notifications to the reporter whenever the status of their reported item changes.
- FR8 – Audit Trail: The system shall maintain a chronological history of all status updates, recording the timestamp and the specific changes made to ensure accountability.

4.4 Non-Functional Requirements

Non-functional requirements (NFRs) define the quality attributes of the system, ensuring it is reliable, efficient, and secure.

- NFR1 – Performance: The application shall render screens at 60 frames per second (FPS) to ensure smooth interaction. Data submission should complete within 3 seconds on standard 4G networks.
- NFR2 – Device Compatibility: The application must be compatible with Android devices running Android 8.0 (Oreo) and above.
- NFR3 – Security: Row Level Security (RLS) shall be enforced at the database layer; role-based access control shall ensure only authenticated admins can modify report statuses or access the admin dashboard.
- NFR4 – Reliability: The application shall handle network interruptions gracefully, displaying appropriate error states and allowing users to retry failed submissions without data loss.

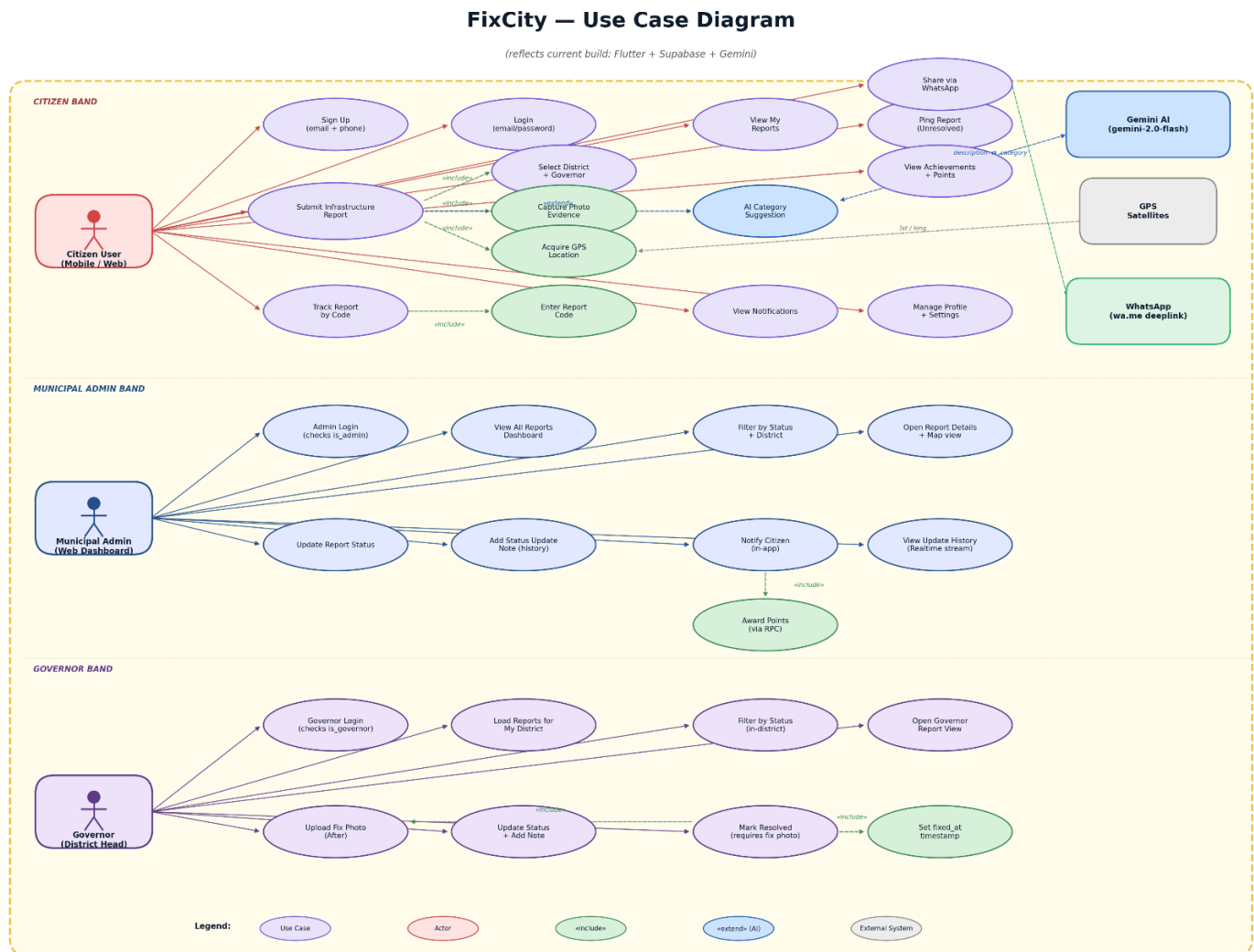
4.5 Use Case Analysis

4.5.1 System Actors

- Citizen (Public User): An authenticated user who has registered via the Sign Up screen. Citizens can submit reports, track their own report status by code, and view their full report history in the My Reports page.

- Admin (Municipal Worker): An authenticated user responsible for reviewing, managing, and resolving reports.

Figure 3: Use Case Diagram



4.5.2 Detailed Use Case Descriptions

Use Case 1: Submit Problem Report

- Actor: Citizen
- Precondition: User has internet access and a compatible device.
- Main Flow:
 - User navigates to the "Submit Report" page.
 - User selects a Category from the dropdown.

- User enters a Description.
- User utilizes the "Get Location" button to auto-fill GPS coordinates.
- User uploads a Photo (optional).
- User clicks "Submit".
- System validates input (e.g., checks if description is empty).
- System saves data to the database with status "New".
- System displays a "Success" message with the Tracking ID.

Use Case 2: Update Report Status

- Actor: Admin
- Precondition: Admin is logged in.
- Main Flow:
 - Admin views the dashboard list.
 - Admin selects a specific report to view details.
 - Admin changes the status dropdown (e.g., from "New" to "In Progress").
 - Admin adds an optional note (e.g., "Maintenance crew dispatched").
 - Admin clicks "Update".
 - System updates the database record.
 - System triggers an in-app notification to the original reporter.

4.6 Data Modeling

The system's data integrity relies on a robust relational database design. The core entities and their attributes are defined below.

4.6.1 Data Dictionary

Entity: auth_users

Description: Managed by Supabase Auth. Stores authenticated user account credentials.

- id (UUID, PK): Supabase Auth UID — unique identifier for every registered user.
- email (String): User's email address.
- last_sign_in (Timestamp): Most recent login timestamp.

Entity: profiles

Description: Stores extended user metadata, gamification data, and role flags.

- id (UUID, PK/FK): References auth_users.id — one-to-one relationship.
- full_name (String): Citizen's display name.
- phone (String): Contact number.
- is_admin (Boolean): Role flag identifying administrator accounts.
- is_governor (Boolean): Role flag identifying governor (district head) accounts.
- governor_district (String): District assigned to this governor.

- points (Integer): Gamification points incremented via RPC.
- sms_enabled (Boolean): SMS notification preference.
- created_at (Timestamp): Account creation date.

Entity: reports

Description: The central entity storing the details of reported infrastructure issues.

- id (UUID, PK): Unique identifier for the report.
- report_code (String): 10-character unique alphanumeric code generated on submission.
- user_id (UUID, FK): References auth_users.id — the user who submitted the report (nullable for anonymous submissions).
- category (String): Problem category key (e.g., cat_pothole, cat_trash, cat_lighting, cat_sewage, cat_water, cat_other).
- description (Text): User-provided details of the issue.
- photo_url (String): URL to the before-photo in Supabase Storage (files/reports/).
- fix_photo_url (String): URL to the after-photo uploaded by the governor upon resolution (files/fix_photos/).
- latitude / longitude (Float): Precise GPS coordinates (Precision 6).
- status (Enum): pending | in_progress | resolved.
- district (String): Administrative district of the report.
- governor_name / governor_phone (String): Assigned governor's contact details.
- ping_count (Integer): Number of times other users have endorsed this report as unresolved.
- fixed_at (Timestamp): Set when status transitions to resolved.
- created_at (Timestamp): Report submission timestamp.

Entity: status_updates

Description: An append-only audit table tracking the history of all changes for each report.

- id (Bigserial, PK): Auto-incrementing log ID.
- report_id (UUID, FK): References reports.id.
- text (Text): The status update note or message added by the official.
- updated_by (String): Identifier for the official making the change ("admin" or "governor").
- updated_at (Timestamp): Timestamp of the update.

Entity: notifications

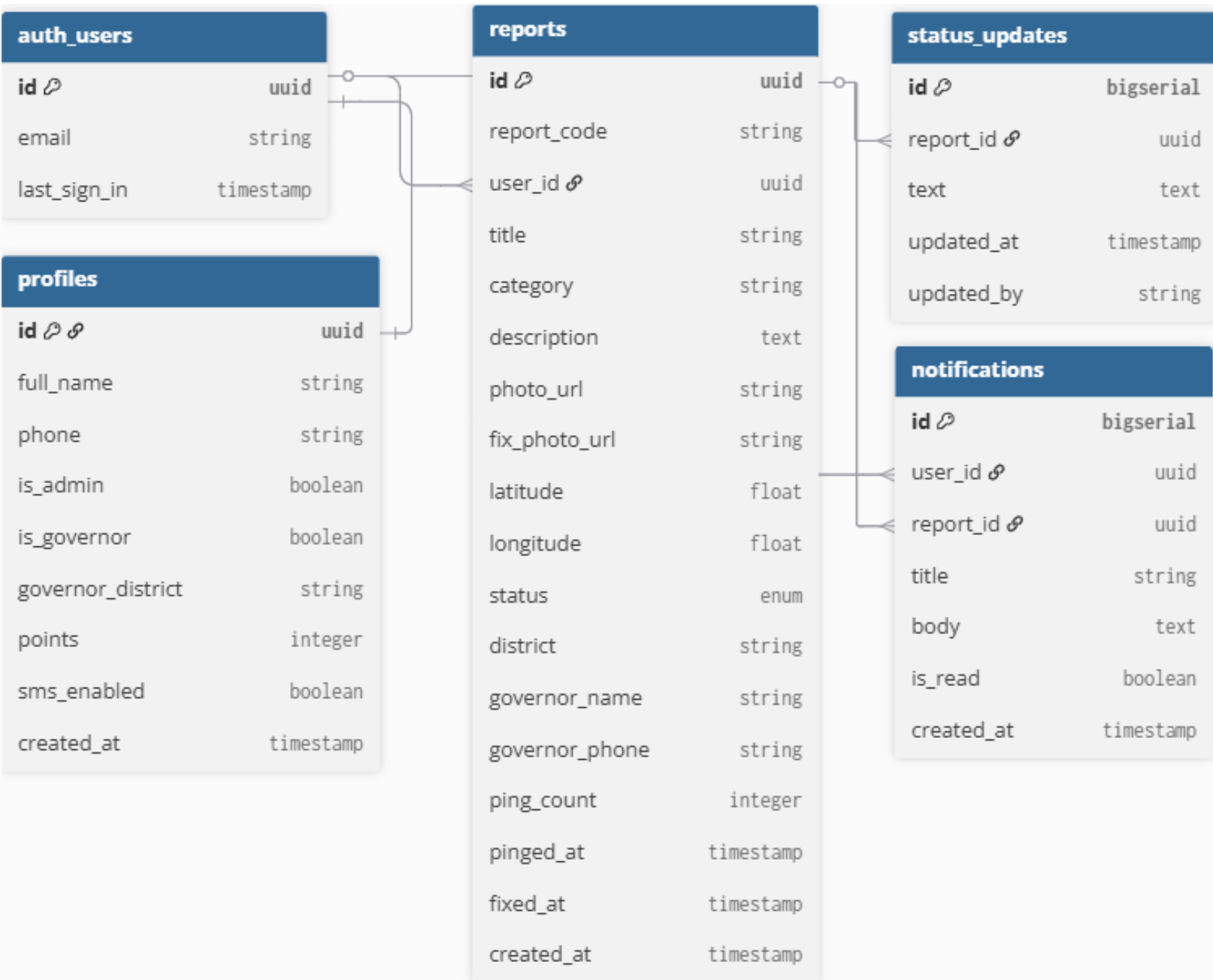
Description: Stores in-app notifications delivered to users on report status changes.

- id (Bigserial, PK): Auto-incrementing notification ID.
- user_id (UUID, FK): References auth_users.id — the recipient of the notification.
- report_id (UUID, Soft FK): References reports.id — the report that triggered the notification (nullable).
- title (String): Notification heading.

- body (Text): Full notification message.
- is_read (Boolean): Read status, default false.
- created_at (Timestamp): Notification creation timestamp.

4.6.2 Entity Relationships

Figure 4: Entity Relationship Diagram (ERD)



- `auth_users` — `profiles`: One-to-One. Every authenticated user has exactly one profile record that stores their role flags and gamification points.
- `auth_users` — `reports`: One-to-Many. A user can submit multiple reports; each report belongs to one user (nullable for anonymous submissions).
- `reports` — `status_updates`: One-to-Many. A single report can accumulate multiple status update entries over its lifecycle.
- `auth_users` — `notifications`: One-to-Many. A user receives multiple notifications across different reports.
- `reports` — `notifications`: One-to-Many. A single report can trigger multiple notifications (e.g., on status change, on resolution).

5. System Design & Implementation

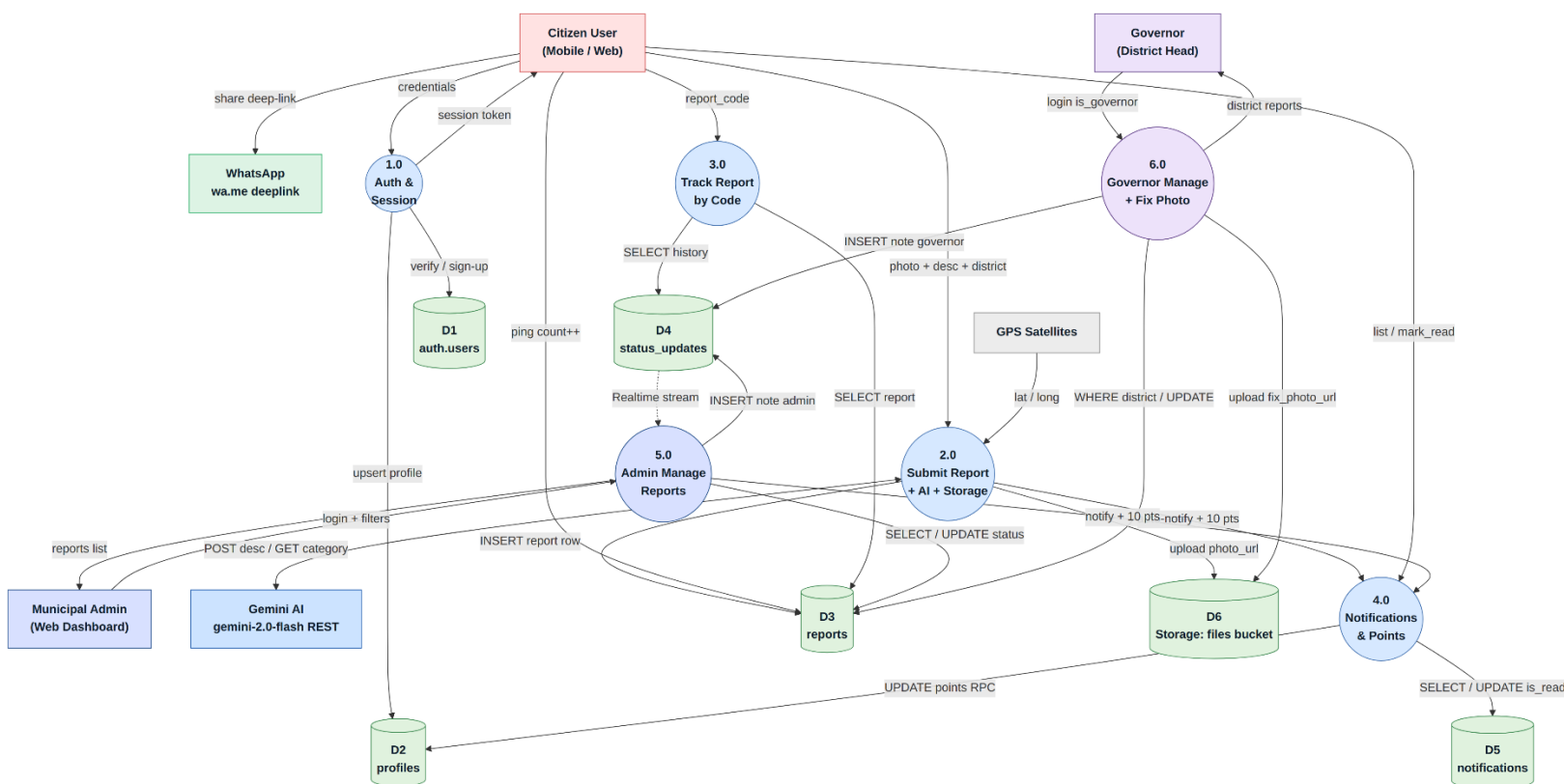
5.1 Introduction

Following the rigorous analysis of system requirements, this chapter details the architectural design and technical implementation of the FixCity platform. It outlines the structural framework of the application, utilizing a layered architecture to ensure scalability and maintainability. The chapter provides an in-depth examination of the User Interface (UI) design principles that guide the citizen experience. Furthermore, it documents the specific technologies employed for development—from the Supabase backend to the Flutter interface—including the AI integration layer using Google Gemini.

5.2 System Architecture

FixCity adopts a modern Client-Server Architecture utilizing a Backend-as-a-Service (BaaS) model.

Figure 5: System Architecture Data Flow Diagram (DFD)



5.2.1 Client Layer (Mobile App)

The client side is a native mobile application built with Flutter.

- **State Management:** The app utilizes local state management to handle user inputs (forms, photos) before submission.

- **UI Components:** Custom widgets are designed using Material Design principles to ensure intuitive navigation.
- **Function:** The app handles GPS data acquisition directly from the device sensors and compresses images before upload to save bandwidth.

5.2.2 Backend Layer (Supabase)

The backend logic is offloaded to the Supabase cloud infrastructure.

- **Database:** A relational PostgreSQL database stores user profiles, report details, and status logs.
- **Storage Buckets:** A dedicated object storage bucket named "files" handles the hosting of evidence photos uploaded by users.
- **Authentication:** Secure user sessions are managed via Supabase Auth (supporting Email/Password logins).

5.2.3 Data Layer (Persistence)

The Data Layer is responsible for the persistent storage and retrieval of all application data, including user credentials, problem reports, and audit logs.

- **Database Engine:** The system utilizes PostgreSQL, hosted on the Supabase cloud platform. PostgreSQL was selected for its robust support for geospatial data, which is essential for storing and querying the location data of infrastructure reports.
- **Data Consistency:** By using PostgreSQL for both development and production environments, the system eliminates compatibility issues, ensuring a seamless deployment pipeline.
- **Data Access:** The application interacts with the database using the Supabase Flutter SDK, performing CRUD operations using secure, parameterized API calls that prevent SQL injection vulnerabilities while mapping database rows directly to Dart objects.

5.2.4 AI Integration Layer (Google Gemini)

A fourth functional layer integrates artificial intelligence into the reporting workflow. The system uses the Google Gemini 2.0 Flash API to automatically classify citizen reports into the appropriate problem category based on natural language descriptions submitted in Arabic or English.

- **Primary Mode:** The citizen types a title and description. The app sends this text to the Gemini API via HTTP POST. Gemini returns one of six category keys (cat_pothole, cat_trash, cat_lighting, cat_sewage, cat_water, cat_other), and the app auto-selects the matching category tile with a sparkle badge indicating AI selection.
- **Fallback Mode:** If the Gemini API is unavailable (CORS restriction on web, network error, or timeout), the system silently falls back to an offline keyword-matching classifier that recognizes problem-related vocabulary in both Arabic and English, ensuring uninterrupted functionality in any environment.

This dual-mode approach was a key differentiator recognized by the Gemini for Graduation Projects program, as it ensures the AI feature delivers value regardless of network conditions — a critical consideration for real-world deployment in areas with variable connectivity.

5.3 User Interface Design

The design philosophy for FixCity prioritizes Simplicity and Accessibility. Given that the primary use case involves users reporting issues while outdoors, the interface must be intuitive and legible under various lighting conditions.

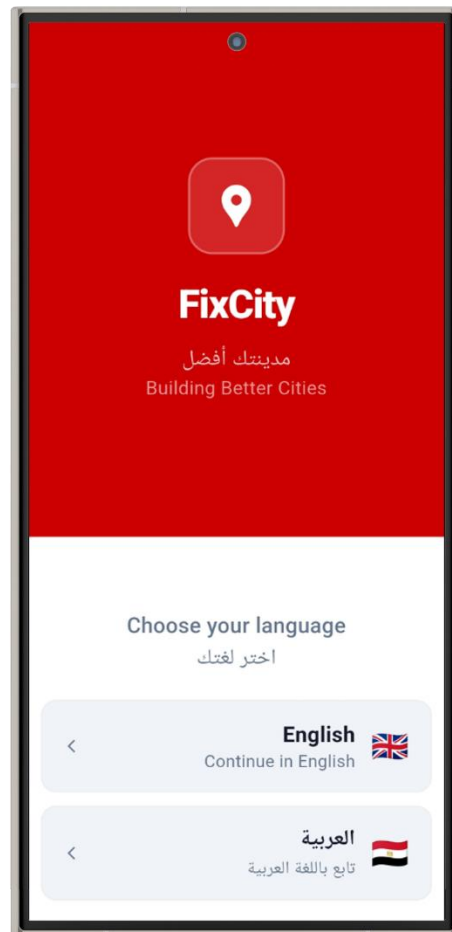


Figure 6: Mobile App Title Screen

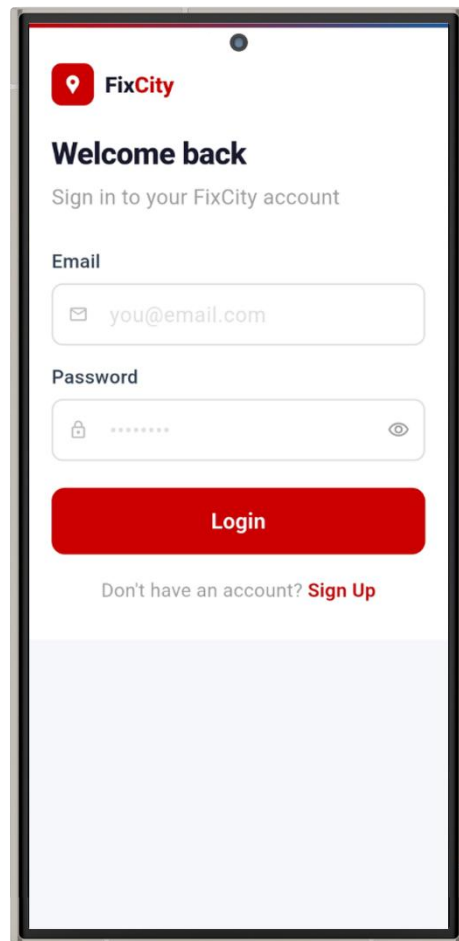


Figure 7: Sign In / Sign Up Screen

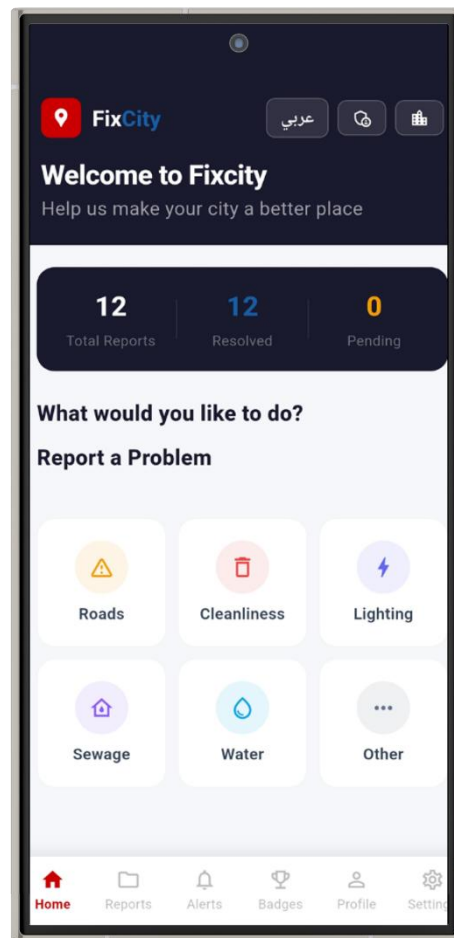


Figure 8: Home Page Dashboard

Submit New Report

Problem Description

Describe the problem in detail...

Suggest Category with AI

Select Category

- Roads
- Cleanliness
- Lighting
- Sewage
- Water
- Other

District

Select a district

Attach Photo

Tap to choose a photo

Set Location My location

Tap the map to pinpoint the exact location

Submit Report

Figure 9: Report Submission Interface

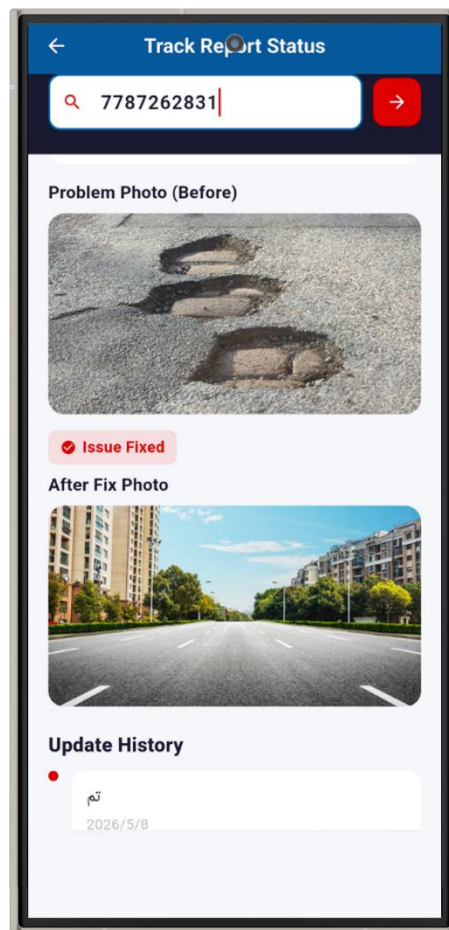


Figure 10: Status Tracking

5.3.1 Homepage & Reporting Form

The homepage serves as the primary entry point. It features a clean header with project branding and a prominent "Submit Report" form.

- **Category Selection:** A dropdown menu allows users to categorize the issue (e.g., "Pothole," "Street Light").
- **Geolocation:** A "Get Location" button utilizes the device's native Geolocation sensors to automatically populate the latitude and longitude fields, reducing user error.
- **Photo Upload:** An integrated file picker allows users to attach photos directly from their device's camera or gallery, with a preview capability to confirm the image before submission.

5.3.2 Submission Confirmation

Upon successful submission, the user is redirected to a confirmation page. This page displays a success message and, crucially, the generated Tracking ID. Clear instructions are provided on how to use this ID to check for future updates.

5.3.3 Administrative Dashboard

The admin dashboard is a restricted area accessible only to municipal workers via secure login.

- **Table View:** It presents a comprehensive table of all submitted reports, sortable by date and status.
- **Filtering:** Admins can filter the view to see only "New" reports or reports specific to a certain category (e.g., "Sanitation").
- **Quick Actions:** Action buttons allow for rapid status updates (e.g., marking a report as "Resolved"), which automatically updates the audit trail.

5.3.4 Status Tracking Page

This public-facing page allows citizens to enter their Tracking ID to view the lifecycle of their report.

- **Visual Timeline:** A graphical timeline displays the report's journey from "Submitted" to "In Review" to "Resolved".
- **Admin Feedback:** Any notes added by municipal workers (e.g., "Scheduled for repair on Monday") are visible here, enhancing transparency.

5.4 Implementation Details

5.4.1 Development Environment

The project was developed using Visual Studio Code equipped with the Dart and Flutter extensions. The application was tested on both Android Emulators and physical Android devices (Samsung Galaxy A346E, Android 14) to ensure hardware compatibility with Camera and GPS features.

5.4.2 Database Schema Implementation

Table 3: Database Schema

Table	Primary Key	Foreign Keys	Key Fields
auth_users	id (uuid)	—	email, last_sign_in
profiles	id (uuid)	id → auth_users.id	is_admin, is_governor, points
reports	id (uuid)	user_id → auth_users.id	report_code, status, category, photo_url
status_updates	id (bigserial)	report_id → reports.id	text, updated_by, updated_at
notifications	id (bigserial)	user_id → auth_users.id, report_id → reports.id	title, body, is_read

Row Level Security (RLS) policies are enforced on all tables. While anyone can insert a report, only authorized admins can update the status field. Storage policies allow public read access for report photos, but upload access is restricted to authenticated sessions.

5.4.3 Security Measures

- Row Level Security (RLS): Logic is enforced at the database layer, ensuring that even if the client app is compromised, unauthorized users cannot modify data.
- Secure API Keys: The application utilizes anonymous public keys for standard requests, while service-role keys are kept strictly on the server side for administrative tasks.

5.5 Testing and Quality Assurance

To ensure system reliability, a structured testing plan was executed. Detailed results are presented in Chapter 6.

- Unit Testing: Individual components, such as the Tracking ID generator and database models, were tested in isolation to verify they function correctly.
- Integration Testing: The interaction between the frontend forms and the backend database was tested to ensure data is correctly captured, stored, and retrieved.
- User Acceptance Testing (UAT): The system was presented to a group of 6 participants representing target demographics to gather feedback on usability and identify workflow bottlenecks.

5.6 Deployment Strategy

- Client Side: The Flutter application is compiled into an APK/AAB file, signed with a release key, and deployed to the Google Play Store for public download. ADB wireless debugging was used during development for physical device testing.
- Server Side: The backend logic is deployed to Supabase Cloud. Database migrations are managed via the Supabase CLI, ensuring that the production database schema remains consistent with the development environment.

5.7 Conclusion and Future Work

FixCity successfully demonstrates that digital transformation in the public sector does not require massive budgets or complex infrastructure. By delivering a functional, user-friendly mobile application, this project provides a blueprint for how Egyptian municipalities can modernize their operations using accessible technology.

Proposed future enhancements include: Advanced Push Notifications via Firebase Cloud Messaging (FCM); GIS Integration providing a visual heat map of infrastructure problems; and AI-powered photo analysis using Gemini Vision to automatically assess damage severity from uploaded photos.

6. Testing & Quality Assurance

6.1 Testing Strategy

The testing strategy for FixCity followed a structured, multi-phase approach. Testing was conducted in two environments: Google Chrome browser via flutter run -d chrome for rapid iteration, and a physical Samsung Galaxy A346E (Android 14) for real-world validation of GPS, camera, and touch interactions.

6.2 Unit Testing

Unit tests verified the correctness of isolated functions and logic components without dependency on external services.

Table 4: Unit Test Results

Test ID	Component	Test Description	Expected Result	Status
UT-01	Report Code Generator	Generate 10 codes, verify uniqueness	All unique, 10 chars, uppercase	PASS
UT-02	AI Keyword Classifier	Input: Arabic pothole text	Returns cat_pothole	PASS
UT-03	AI Keyword Classifier	Input: "broken street light"	Returns cat_lighting	PASS
UT-04	AI Keyword Classifier	Input: Arabic water leak text	Returns cat_water	PASS
UT-05	AI Keyword Classifier	Input: "unrelated text xyz"	Returns cat_other	PASS
UT-06	Form Validation	Submit with empty description	Prevents submission, error shown	PASS
UT-07	Form Validation	Submit with no category selected	Prevents submission, error shown	PASS
UT-08	Form Validation	Submit with no photo attached	Prevents submission, error shown	PASS
UT-09	Language Toggle	Switch Arabic to English	LTR layout, labels in English	PASS
UT-10	Language Toggle	Switch English to Arabic	RTL layout, labels in Arabic	PASS
UT-11	Problem.fromSupabase()	Parse Supabase JSON response	All fields correctly mapped	PASS
UT-12	Status Color Mapping	Map pending to expected color	Returns amber Color(0xFFFF59E0B)	PASS

6.3 Integration Testing

Integration tests validated the correct interaction between the Flutter frontend and the Supabase backend across the full system stack.

Table 5: Integration Test Results

Test ID	Flow	Test Steps	Expected Result	Status
IT-01	Citizen Registration	Sign up with new email/password	User in Auth, profile created	PASS
IT-02	Citizen Login	Login with valid credentials	Session created, go to HomePage	PASS
IT-03	Admin Login (Valid)	Login with is_admin=true account	Redirected to AdminDashboardPage	PASS
IT-04	Admin Login (Invalid)	Login with citizen at admin page	Session revoked, Access Denied	PASS
IT-05	Report Submission	Complete form and submit	Row in DB, photo in storage, code	PASS
IT-06	Report Tracking (Valid)	Enter valid report code	Report details and timeline shown	PASS
IT-07	Report Tracking (Invalid)	Enter non-existent code	Error: Report not found	PASS
IT-08	Admin Status Update	Add update, change to resolved	DB updated, audit entry created	PASS
IT-09	My Reports	Authenticated citizen opens page	Only user's own reports shown	PASS
IT-10	Live Stats	Submit report, refresh home	Count incremented correctly	PASS
IT-11	Gemini AI (Mobile)	Type description, tap AI button	Category auto-selected from Gemini	PASS
IT-12	AI Fallback (Chrome)	Same test on Chrome browser	Keyword classifier activates	PASS
IT-13	Photo Upload	Attach 2MB JPEG to report	Photo in Storage, URL in reports	PASS
IT-14	Map Interaction	Tap different location on map	Marker moves, coordinates updated	PASS

6.4 User Acceptance Testing (UAT)

6.4.1 UAT Participants

UAT was conducted with 6 participants representing the target user demographics. Participants completed standardized tasks without assistance.

Table 6: UAT Participants

Participant	Age	Technical Level	Device
P1	22	High (Student)	Samsung Galaxy A34
P2	45	Low (Non-tech professional)	iPhone 13
P3	31	Medium (Office worker)	Xiaomi Redmi Note 12
P4	19	High (IT Student)	Samsung Galaxy S23
P5	55	Low (Retired)	Huawei Y9
P6	28	Medium (Engineer)	OnePlus Nord

6.4.2 UAT Task Results

Table 7: UAT Task Results

Task	Description	Completion Rate	Avg. Time	Notes
Task 1	Select language and proceed to home	6/6 (100%)	18 sec	All completed without assistance
Task 2	Submit report with photo and location	6/6 (100%)	2 min 14 sec	P2 needed extra time on photo button
Task 3	Use AI category suggestion button	5/6 (83%)	45 sec	P5 did not understand AI button initially
Task 4	Track a report using provided code	6/6 (100%)	38 sec	All completed quickly
Task 5	Switch language mid-session	6/6 (100%)	12 sec	RTL/LTR switch worked correctly for all
Task 6	Locate and open My Reports page	5/6 (83%)	1 min 2 sec	P5 initially confused with Track Reports

6.4.3 UAT Findings & Actions

- Positive: 100% of participants submitted a report within 3 minutes, validating the UI simplicity goal.
- Positive: AI suggestion button was well-received by technical users, described as impressive and intuitive.
- Action Taken: AI button label updated with sparkle emoji to communicate its AI nature more clearly.
- Action Taken: Three-page onboarding sequence added to orient first-time users before reaching the home screen.
- Positive: Language switching was universally successful across all tested devices and screen sizes.

6.5 Device Compatibility Testing

Table 8: Device Compatibility Testing Results

Device	Android Version	Screen Size	Notch Type	Result
Samsung Galaxy A346E	Android 14	6.4 inch	Punch-hole	PASS – primary test device
Samsung Galaxy S23	Android 14	6.1 inch	Punch-hole	PASS
Xiaomi Redmi Note 12	Android 13	6.67 inch	Teardrop	PASS
Huawei Y9	Android 9	6.5 inch	Dewdrop	PASS
Google Chrome (PC)	N/A	Various	N/A	PASS – AI uses keyword fallback on web

6.6 Performance Testing

Performance metrics were recorded on the primary test device (Samsung Galaxy A346E) under normal WiFi conditions.

Table 9: Performance Testing Results

Metric	Target	Measured Result	Status
Cold app launch time	< 3 sec	1.8 sec	PASS
Report submission (with photo)	< 10 sec	4.2 sec avg	PASS
Map tile loading	< 3 sec	2.1 sec (WiFi)	PASS
Report tracking query	< 2 sec	0.9 sec avg	PASS
Home stats loading	< 2 sec	1.1 sec avg	PASS
Admin dashboard (20 reports)	< 3 sec	1.4 sec	PASS
AI classification (Gemini API)	< 8 sec	2.3 sec avg	PASS
AI fallback (keyword matching)	< 0.1 sec	< 0.01 sec	PASS

6.7 Security Testing

- Unauthorized Admin Access: Direct navigation to /admin/dashboard without auth redirects to login — PASS
- Citizen Admin Attempt: Citizen account at admin login is immediately signed out with Access Denied — PASS
- Data Isolation: My Reports only returns the authenticated user's own reports — PASS
- RLS Enforcement: Direct API calls attempting to UPDATE status without admin credentials rejected with 403 — PASS
- API Key Security: No service-role keys found in client source code — PASS

7. Conclusion & Future Work

7.1 Project Summary

This project addressed a well-documented but largely unresolved challenge in Egyptian urban governance: the inefficient and inaccessible communication channel between citizens and municipal authorities for infrastructure problem reporting. The result is FixCity, a fully functional, bilingual, AI-enhanced mobile platform that democratizes the reporting process and provides municipal workers with a structured, real-time management tool.

The platform was accepted into the Gemini for Graduation Projects program (Cohort III) by Gemini Africa, in collaboration with the Academy of Scientific Research and Technology — recognition that validates both the technical quality and the social impact potential of the solution.

7.2 Achievement of Objectives

Table 10: Achievement of Objectives

Objective	Achievement	Status
Native mobile app for geo-tagged reports	Flutter app with GPS map picker and camera, deployed to Android	Achieved
Unique tracking code per report	10-character alphanumeric code generated on submission	Achieved
AI-powered category classification	Google Gemini 2.0 Flash API with offline keyword fallback	Achieved
Role-based admin dashboard	Admin panel with RLS, filters, status management, real-time updates	Achieved
Bilingual Arabic/English support	Full RTL/LTR dynamic switching with custom translation map	Achieved
Security and data access control	Supabase RLS + is_admin role check on all admin routes	Achieved
Deploy to physical Android devices	Signed APK tested on Samsung A346E and four additional devices	Achieved

7.3 Research Contributions

- **BaaS for civic technology:** Demonstrates enterprise-grade features without dedicated server infrastructure, a reusable pattern for resource-constrained public sector projects.
- **Dual-mode AI classification:** Cloud AI with offline keyword fallback provides a reliable AI integration pattern for applications in varying network conditions.
- **Bilingual Flutter architecture:** Dynamic RTL/LTR implementation demonstrates a practical approach to Arabic/English bilingual mobile applications.
- **Egyptian municipal technology blueprint:** A cost-effective model for modernizing urban management workflows in small-to-medium Egyptian municipalities.

7.4 Limitations

- No Push Notifications: Citizens must actively open the app to check status; Firebase Cloud Messaging is planned for Phase 2.
- No Offline Support: Requires active internet connection; offline draft-with-later-sync is planned for Phase 2.
- AI Web Limitation: Gemini API is blocked by CORS in Chrome; keyword fallback activates seamlessly but Gemini classification is Android-only.
- Single Municipality Scope: Current architecture supports one municipality; multi-tenant support requires database restructuring.
- Manual Admin Account Creation: Admin accounts must be created manually in Supabase; no self-service invitation flow exists.

7.5 Future Work

7.5.1 Phase 2 Features (Short-Term)

- Push Notifications: Firebase Cloud Messaging for instant citizen alerts on status changes and admin notifications.
- AI Photo Analysis: Extend Gemini to analyze uploaded photos using Gemini Vision for damage classification.
- GIS Heat Map Dashboard: Admin-facing map view showing geographic distribution of reports.
- Offline Mode: Local SQLite caching for offline report drafting with automatic submission on reconnect.

7.5.2 Phase 3 Features (Long-Term)

- Multi-Municipality Support: Multi-tenant architecture for different municipalities with isolated data environments.
- Municipality ERP Integration: REST API layer enabling FixCity to push data into existing municipal systems.
- Advanced Analytics Dashboard: Data visualization showing resolution times, hotspots, and seasonal trends for municipal administrators.
- iOS Support: Expansion of the Flutter application to iOS with App Store deployment.

7.6 Final Conclusion

FixCity demonstrates that meaningful digital transformation in public services does not require massive budgets, proprietary infrastructure, or years of development. By combining Flutter's cross-platform capabilities, Supabase's zero-infrastructure backend, and Google Gemini's artificial intelligence, the team delivered a production-ready platform that is functional, secure, bilingual, and genuinely useful.

The platform directly addresses the Digital Divide identified in Chapter 1, where national digital transformation goals have not yet reached the last mile of municipal management. Every citizen

with a smartphone can now report an infrastructure problem in under 3 minutes, in their preferred language, with GPS precision and photo evidence. Every municipal worker can manage, respond to, and resolve those reports from a clean, mobile-first dashboard.

The acceptance of this project into the Gemini for Graduation Projects program validates its real-world applicability and positions FixCity not merely as an academic exercise, but as a genuine foundation for civic technology in Egypt. In alignment with the United Nations Sustainable Development Goals, specifically SDG 11 (Sustainable Cities and Communities) and SDG 16 (Peace, Justice and Strong Institutions), FixCity represents a meaningful, scalable, and technically sound contribution to Egypt's journey toward a smarter, more transparent, and more accountable urban future.

8. References

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9. Appendices

Appendix A: User Survey Questionnaire and Results Summary

A quantitative survey was conducted to validate the problem statement and assess the market readiness for a mobile reporting solution. The survey received 167 responses from urban residents.

Key Findings Summary:

- **Public Need:** 100% of respondents stated they have faced public infrastructure issues (broken lights, road damage, waste).
- **Current Behavior:** The "Silent Majority" phenomenon was confirmed, with 88.2% of respondents indicating they usually "Do not report" issues due to the lack of accessible channels.
- **Device Preference:** A significant majority (94.1%) prefer using a Mobile Phone for reporting over computers or laptops.
- **Solution Acceptance:** 100% of participants confirmed they would use an online platform like FixCity if it were available.

Would you use an online platform like FixCity to report city problems?

 [Copy chart](#)

167 responses

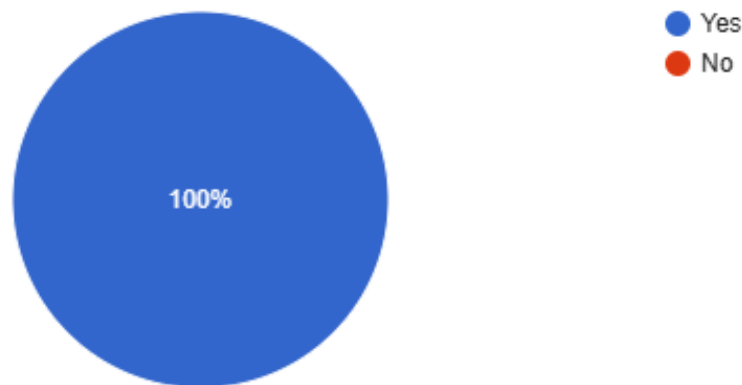
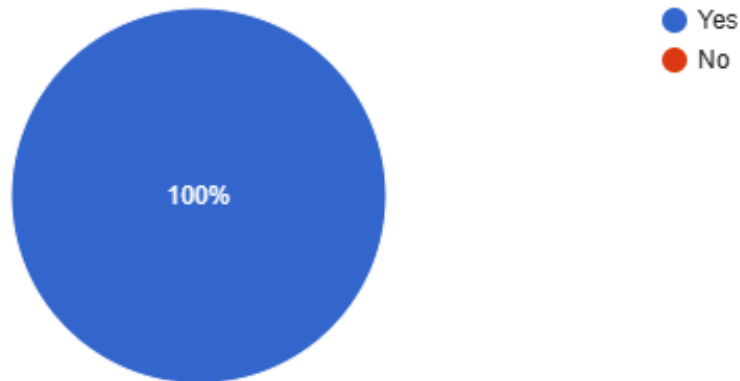


Figure 11: Survey Response – User Preference on Application Type

Have you ever faced public issues such as broken lights, road damages, or waste problems?

 [Copy chart](#)

167 responses



how do you usually report such issues?

 [Copy chart](#)

167 responses

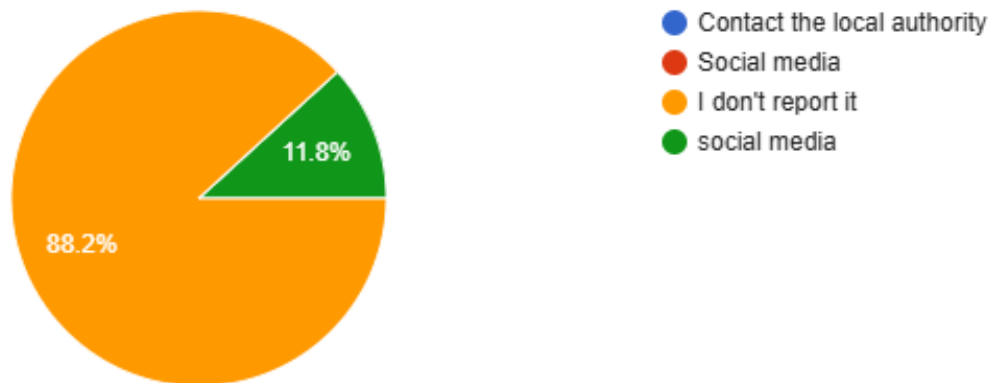
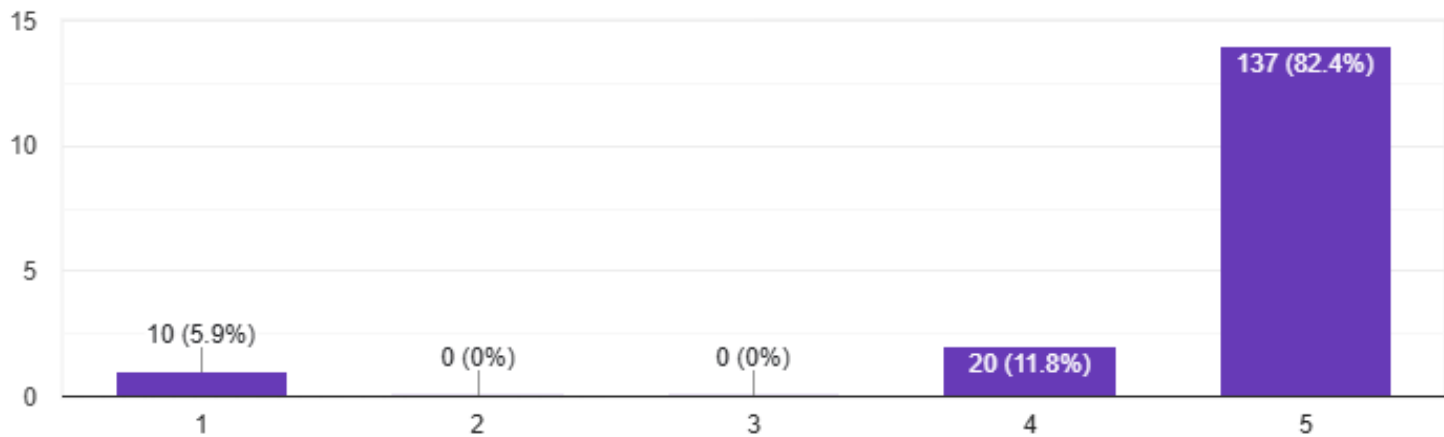


Figure 12: Survey Response – User Preference on Device for reporting

How do you think such a system is important for Egypt's cities?

 Copy chart

17 responses



What device would you prefer to use for reporting ?

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167 responses

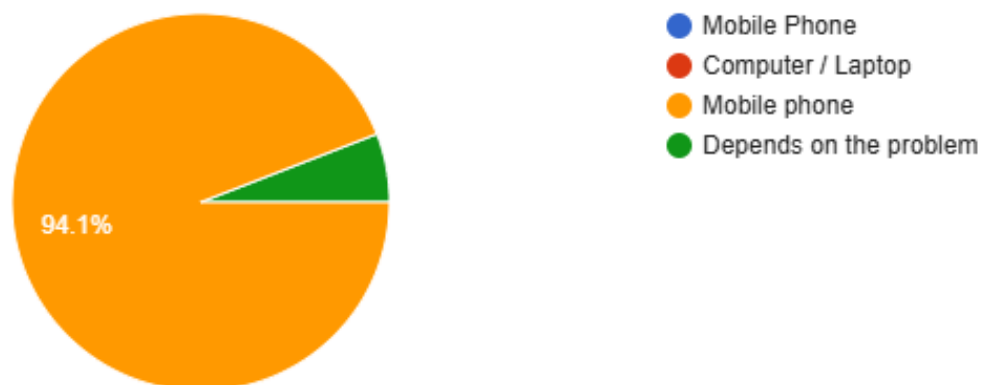


Figure 13: Survey Response – User Experience on Infrastructure Problems & Reports

Appendix B: System Implementation Timeline and Task Breakdown

Based on the phased implementation strategy outlined in Chapter 3, the following timeline projects the rollout of the FixCity platform over a 12-month period.

Table 11: System Implementation Timeline (Phased Rollout)

Phase	Duration	Key Activities	Deliverables
Phase 1: Awareness	Months 1–3	Finalizing MVP development; presentations to local council members; digital social media outreach	Fully tested APK; Project Documentation; Marketing Materials
Phase 2: Pilot	Months 4–6	Beta deployment in one partner municipality; training municipal staff; collecting real-world user feedback	Pilot Case Study; Performance Metrics Report; First 500 Active Users
Phase 3: Expansion	Months 7–12	Regional rollout to adjacent districts; integration of advanced features (Push Notifications); partnership with local NGOs	Version 2.0 Release; 5,000 Active Users; Revenue Generation

Appendix C: Database Schema Detailed Design

The backend utilizes a relational PostgreSQL database hosted on Supabase. The following details the core table structures utilized in the MVP.

1. Table: profiles

- `id` (UUID, PK): Links to Supabase Auth ID.
- `full_name` (Text): Citizen's display name.
- `is_admin` (Boolean): Role flag identifying administrator accounts.
- `is_governor` (Boolean): Role flag identifying governor accounts.
- `created_at` (Timestamp): Account creation date.

2. Table: reports

- `id` (UUID, PK): Unique identifier for the report.
- `user_id` (UUID, FK): Reference to the submitting user (`auth_users.id`).
- `category` (Text): Problem category key (e.g., `cat_pothole`, `cat_trash`, `cat_lighting`).
- `description` (Text): User-provided details.
- `latitude/longitude` (Float): Precise GPS coordinates (Precision 6).
- `photo_url` (Text): URL to the report image in Supabase Storage.
- `fix_photo_url` (Text): URL to the resolution image uploaded by the governor.
- `status` (Enum): `pending` | `in_progress` | `resolved`.

3. Table: status_updates

- id (Bigserial, PK): Auto-incrementing log ID.
- report_id (UUID, FK): The report being updated.
- updated_by (Text): Identifier for the official making the change ("admin" or "governor").
- text (Text): The status update note or message added by the official.

Appendix D: User Interface Mockups and Wireframes

The design process utilized Figma to create high-fidelity prototypes prior to Flutter development. This ensured the "Simplicity and Accessibility" philosophy was met before writing a single line of code.

Interactive Prototype: [View on Figma](#)

Key Screens Included:

- Authentication: Login/Sign-up flows supporting secure access.
- Home Dashboard: Quick access to "Submit Report" and "Track Status".
- Reporting Interface: GPS auto-detection and Camera integration.
- Admin Panel: Map-based filtering of active city issues.

Appendix E: Source Code Documentation and Repository Information

The complete source code for the FixCity mobile application (Flutter) and backend configuration (Supabase) is hosted in a public repository.

Repository URL: <https://github.com/moatazahmedshaker/Fixcity>

- Platform: GitHub
- Technologies: Dart, Flutter, SQL (PostgreSQL)

Appendix F: Testing Plan and Quality Assurance Procedures

To ensure the reliability of the MVP, the following testing strategies were executed:

1. Unit Testing

- Scope: Individual Dart functions and logic components.
- Key Tests: Validation logic for description fields; GPS coordinate parsing reliability; Tracking ID generation uniqueness.

2. Integration Testing

- Scope: Interaction between Flutter Frontend and Supabase Backend.
- Key Tests: Auth Flow — verifying successful login returns a valid Session Token; Data Persistence — confirming a submitted report appears in the PostgreSQL database; Storage — verifying images are correctly uploaded to the bucket and linked to the report row.

3. User Acceptance Testing (UAT)

- Scope: Usability testing with non-technical users.
- Method: A group of 6 users were asked to complete defined tasks without assistance.
- Result: All users completed the core report submission task in under 3 minutes, validating the UI design.